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PI Name:	Clark, Noel A. Ph.D.		
Project Title:	Structure and Dynamics of Freely Suspended Liquid Crystals		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	98-HEDS-03
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note - end date changed from 11/30/2003 grant renewed for 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Maclennan, Joseph Edward (University of Colorado) Glaser, Matthew Arthur (University of Colorado)		
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We pursue ground-based studies of freely suspended liquid crystal films with an eye toward µg experiments, as well as atomistic computer simulation of liquid crystal phases, probing the molecular origins of observed film behavior. Ground-based studies focus on tethered smectic bubbles on which islands (isolated pancake-shaped thicker regions) can be generated and observed. The islands form a novel interacting particle array confined to the two-dimensional surface of the bubble. We observe islands (using reflected light microscopy) and manipulate them (using optical tweezers) and carry out computer simulation of the interaction of islands on contact, in an effort to understand their stability against aggregation. Our work is directed toward developing a deeper understanding of condensed phases under conditions of reduced dimensionality. Films are locally optically anisotropic so that the pattern of molecular orientation within the film plane can be effectively studied by depolarized reflected light microscopy are also being studied. In particular,

Task Description:	topological defect motion can be observed and the simplest liquid crystal defect dynamical process, the attraction and annihilation of oppositely charged, unit-strength defects (vortex-antivortex pairs), probed. Experiments to date reveal significant shortcomings in our understanding of this process. Application of in-plane electric fields to tilted smectic films reveals a host of interesting field-induced structures and transitions, and leads to novel methods for probing film structure. For example, by comparing the response of even layer numbered and odd layer numbered films, basic information about the nature of the polar ordering (ferroelectric or antiferroelectric) of the smectic can be obtained. Our recent discovery of the spontaneous formation of chiral domains in fluid smectic phases of achiral banana-shaped molecules (the first observation of spontaneous chiral domain formation in a fluid since Pasteur's discovery in crystals 150 years ago), based on observation of free film response to applied electric field, will be pursued, as will the recent discovery of freely suspended filaments of smectics comprised of banana-shaped molecules. Smectic liquid crystals are phases of rod shaped molecules organized into one dimensionally (1D) periodic arrays of layers, each layer being a two dimensional (2D) liquid between one and two molecular lengths thick. This inherent fluid-layer structure and the low vapor pressure of smectic liquid crystals enable the long term stabilization of freely suspended, single component, fluid films quantized in integral numbers of layers in thickness, down to a single molecular layer (30Å). These ultrathin freely suspended liquid crystal films are structures of fundamental interest in condensed matter and fluid physics. They are the thinnest known stable condensed phase fluid structures and have the largest surface-to-volume ratio of any stable fluid preparation, making them ideal for the study of the effects of reduced dimensionality on phase behavior and on fluctuation and interface phenomena. Their low vapor pressure and quantized thickness enable the effective use of microgravity to extend the study of basic capillary phenomena to ultrathin fluid films.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	INTRODUCTION: Smectic liquid crystals composed of fluid layers can be made to form freely suspended, single component, fluid films as thin as a single molecular layer (3nm). These films are of fundamental interest in condensed matter physics. They are the thinnest known stable fluid structures and have the largest surface-to-volume ratio of any fluid preparation, making them ideal for studies of two dimensional phase transitions, fluid flow, and fluctuation and interface phenomena. They offer unique opportunities to effectively use microgravity to extend the study of basic capillary phenomena to ultrathin fluid films. The smectic layering forces films to be an integral number of layers thick, quantizing their thickness in layer units and forcing a film of a particular number of layers N to be physically homogeneous with respect to its layer structure over its entire area. This inherent fluid-layer structure and the low vapor pressure of smectics enable, in a straightforward way, the stabilization of freely suspended, single component, layered fluid films for detailed study. The main objective of the research supported by this Grant is to advance the understanding of fluid physics by exploiting the unique characteristics of freely suspended liquid crystal films in a microgravity environment. These experiments focus on use of a novel collective system of interacting, one dimensional interfaces in two dimensional fluids for the study of the fluid physics in two dimensions and the capillarity of ultrathin fluid films. We propose to pursue the definition of experiments on island generation, observation (using reflected light microscopy), and manipulation (using optical tweezers) for the µg environment. PROGRESS: The past year has seen the following progress: We have developed bubble tethering on razor sharp needles providing the ultraweak coupling of the film to the bulk fluid, as required for the island interaction studies we are carrying out. We have made operational out mulitrap optical tweezer system and are applying it to various liquid crystal problems, in particular trapping of islands on freely suspended films. The trap geometry is dynamically reconfigurable using a Labview program we developed. We have directly measured the elastic dipolar interactions between islands on freely suspended smectic films using multitrap optical tweezer system. We have observed the very interesting trapping behavior and hydrodynamic convective motions in our optical trapping experiments using colloidal dispersions of micron-sized silica and polystyrene spheres in a nematic liquid crystal E31 and 5CB and in a semectic liquid crystal 8CB. PROPOSED PLANS: Studies of trapping behavior and convective hydrodynamic motion in nematic colloids using optical tweezers - Interactions between solid particles dispersed in a nematic fluid with hyperbolic hedgehog defects have been studied by using a laser trapping method. In our study, we propose to extend optical trapping experiments to look at the rotating behavior of an optically trapped particle in circularly polarized light, stable particle arrangement and artificial construction of multi-particle regular arrangements in nematic and smectic colloids. Also, we have demonstrated the hydrodynamic toroidal convective motions in our optical trapping experiments using colloidal dispersions of micron-sized silica in a nematic liquid crystal 5CB. The cell is locally heated either from top or from bottom by Gaussian laser radiation and hydrodynamic convective motion are conditioned mainly by thermal gradients. We will develop the model to explain our laser-induced hydrodynamic convective motion in nematic colloids which can be considered as a classical Raleigh-Benard problem of a liquid crystal layer of thickness L between horizontal plates with a vertical temperature change (dT/dz) and a horizontal change (dT/dr) determined by the Gaussian beam profile. Island Coalescence Study - The set-up to measure number of layers in the bubble and freely suspended films by laser reflectivity will be incorporated into the microscope with laser tweezers. In combination with laser reflectivity and light spectroscopy, number of layers of thick bubble and island can be very accurately measured. Using the previously developed laser tweezing equipment for manipulating islands, we will study coalescence induced by pushing islands into one another and also compare natural coalescence with induced coalescence. In this experiment, the interaction forces between islands will be also measured as a function of distance, thickness, and temperature in the smectic A and C liquid crystal phases. Isotropic droplets on Freely Suspended Films - We propose to extend optical trapping experiments to look at the behavior of droplets of isotropic fluids, such as glycerol and water, which can be sprayed onto the films using inkjet printer technology. In this experiment, the interactions between isotropic droplets on anisotropic liquid crystal films, chain formation of droplets, and artificial construction of droplet arrangements using optical tweezers will be studied.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
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Presentation	Pattanaporkratana, A.; Park, C. S.; Maclennan, J. E.; Clark, N. A. "MANIPULATION OF ISLANDS ON FREELY SUSPENDED SMECTIC FILMS AND BUBBLES USING OPTICAL TWEEZERS" 20th International Liquid Crystal Conference, Ljubliana, Slovenia Jul-2004